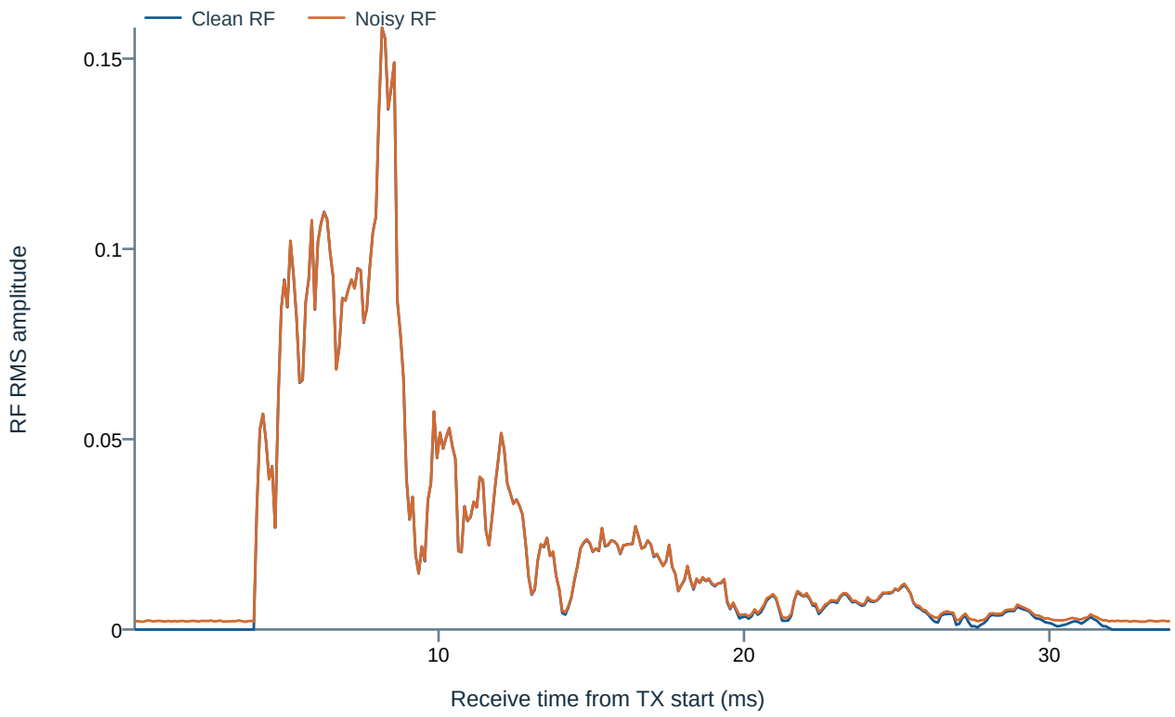
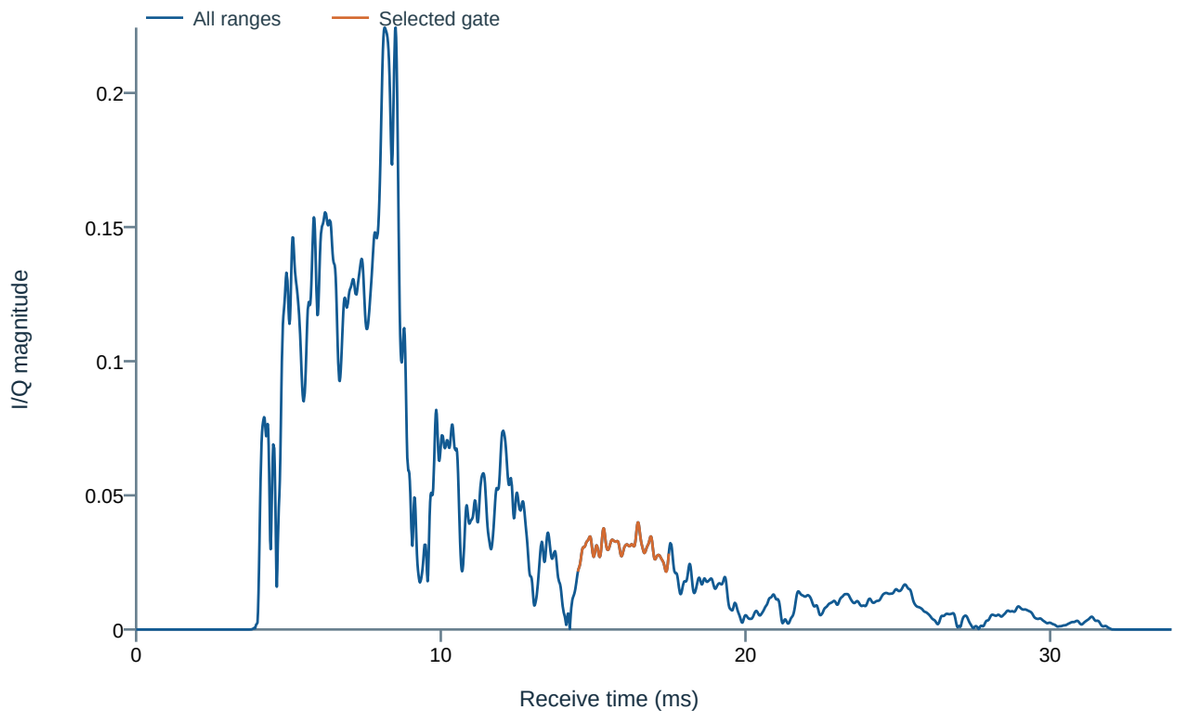
1 / Full received record, then one gate

1440 scatterers in 3-21 m | 4 ms narrowband pulse | Real RF is summed before demodulation

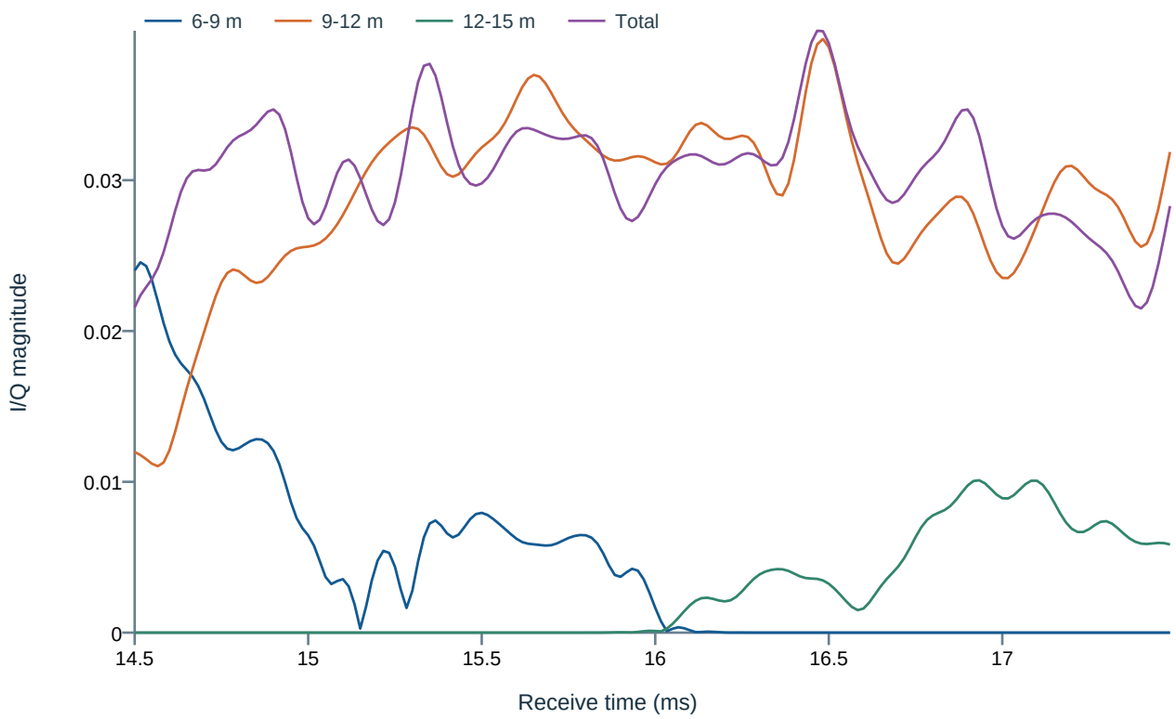
A. Whole-record RF amplitude (0.1 ms RMS)



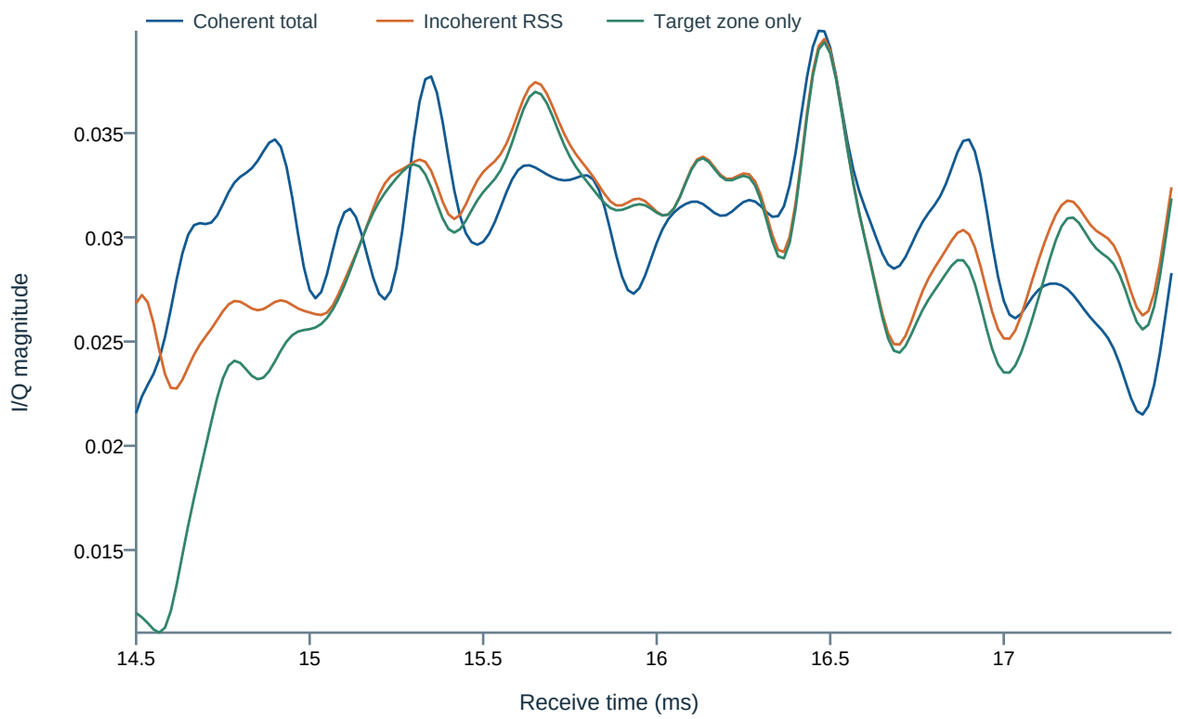
B. Full I/Q envelope and selected gate



C. Several range zones enter ONE gate



D. Contributions can reinforce or cancel

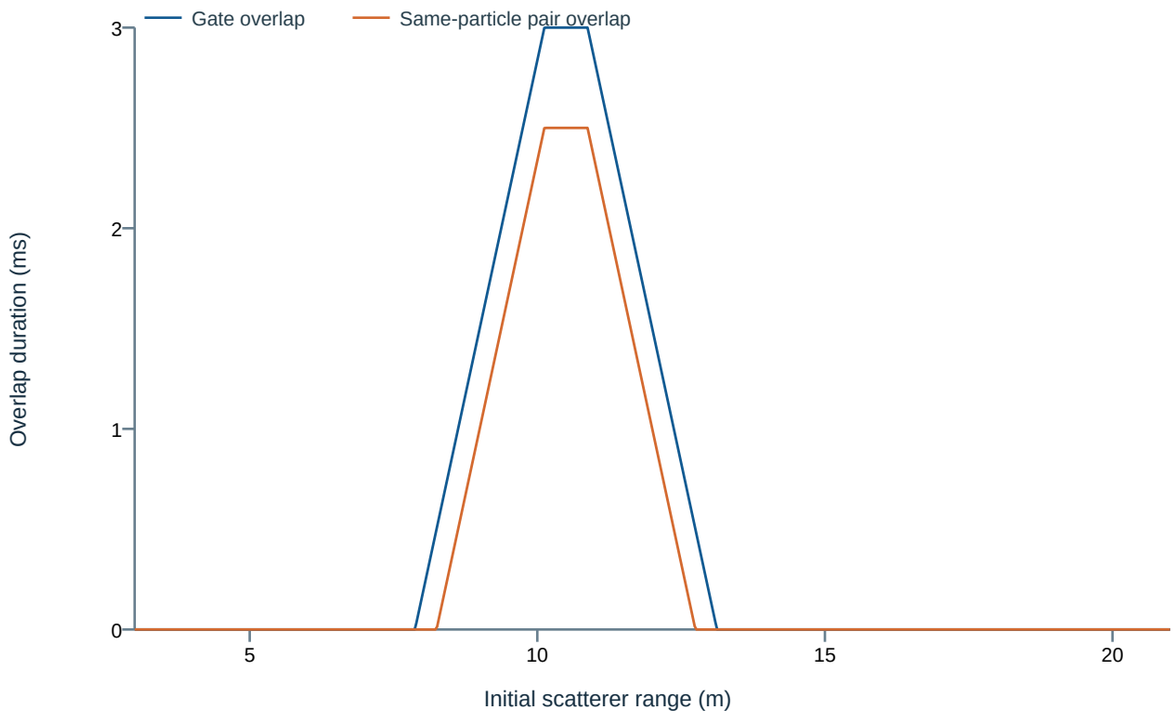


Narrowband teaching simulation | Positive radial velocity = away | Offline FIR delay compensated

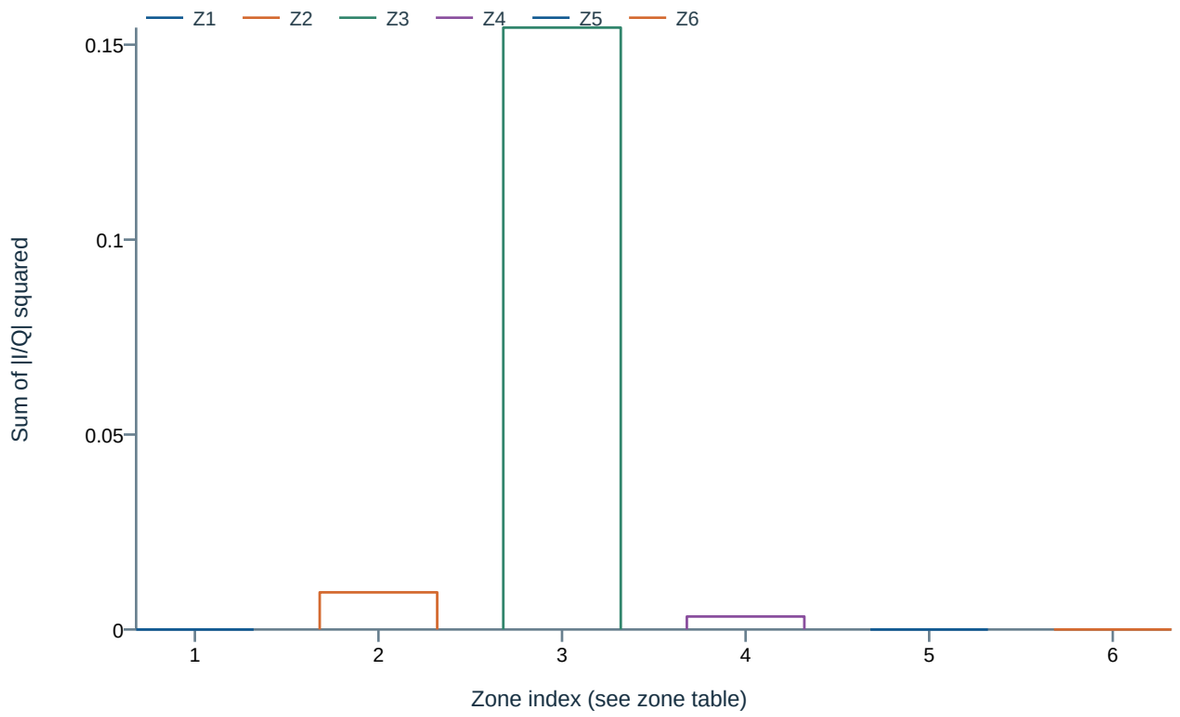
2 / Which ranges influence this gate?

Gate = 14.5-17.5 ms | Nominal range center = 10.5 m | No multipath or previous-ping echoes

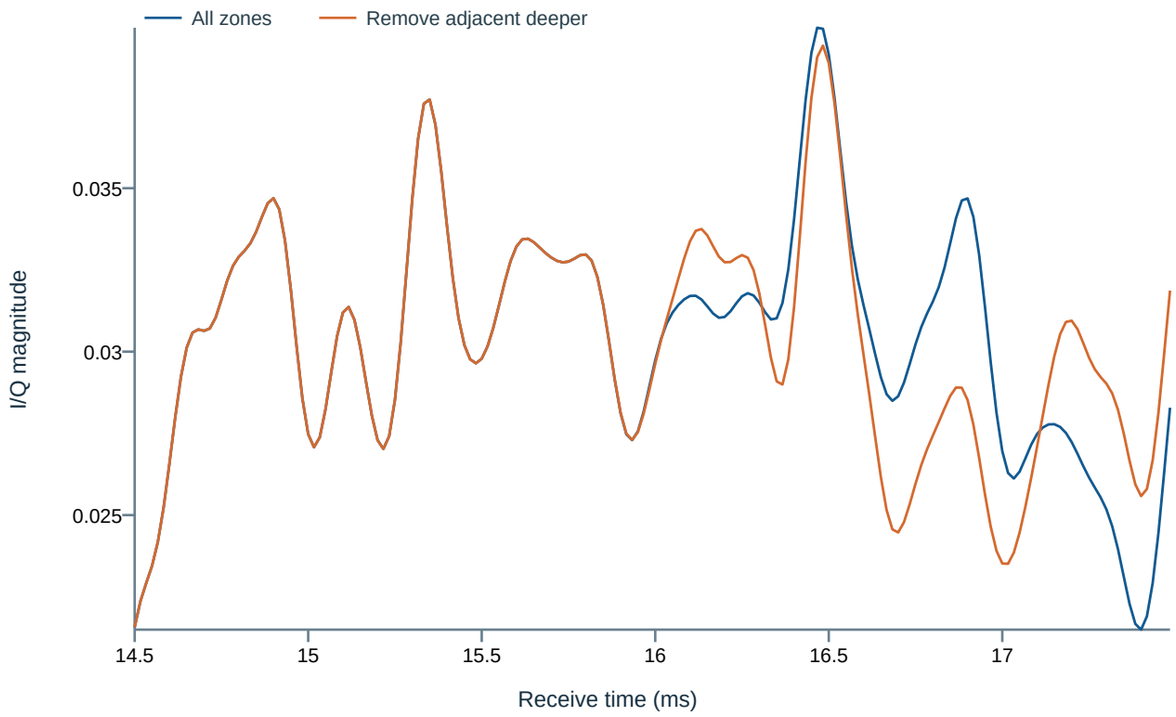
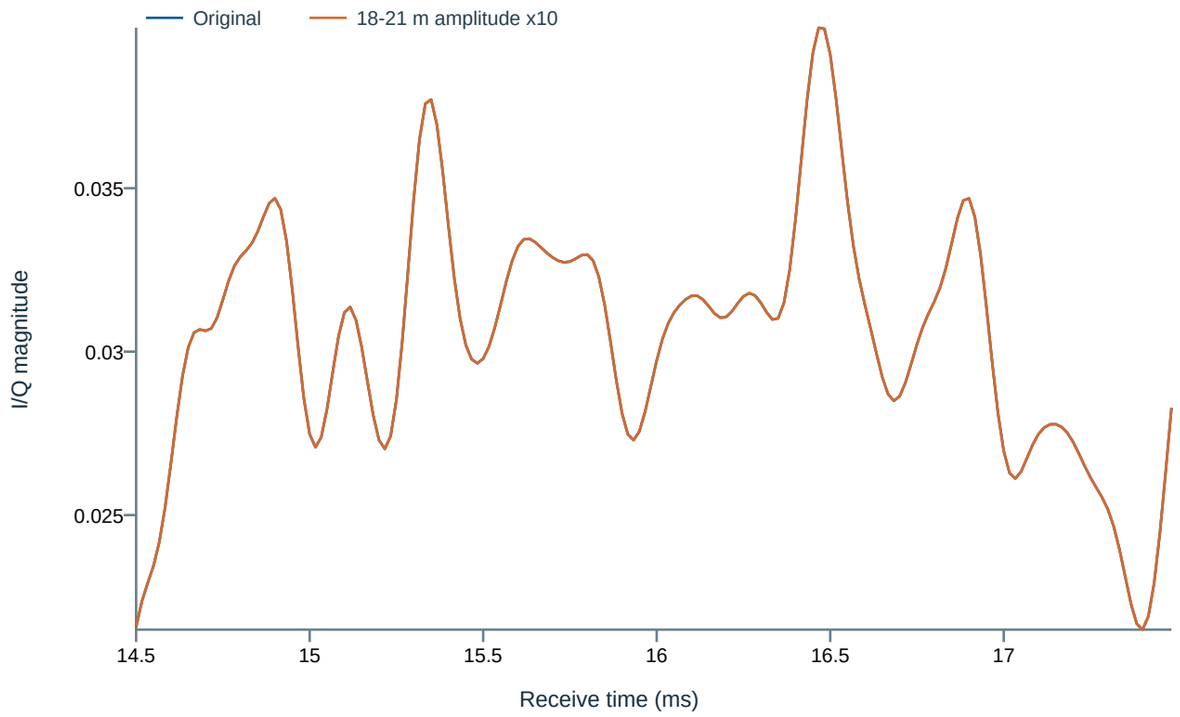
A. Geometric contribution vs initial range



B. Per-zone gate energy (before cross terms)



C. Remove the 12-15 m zone



D. A distant direct echo arrives too late